1. create file, load CSV

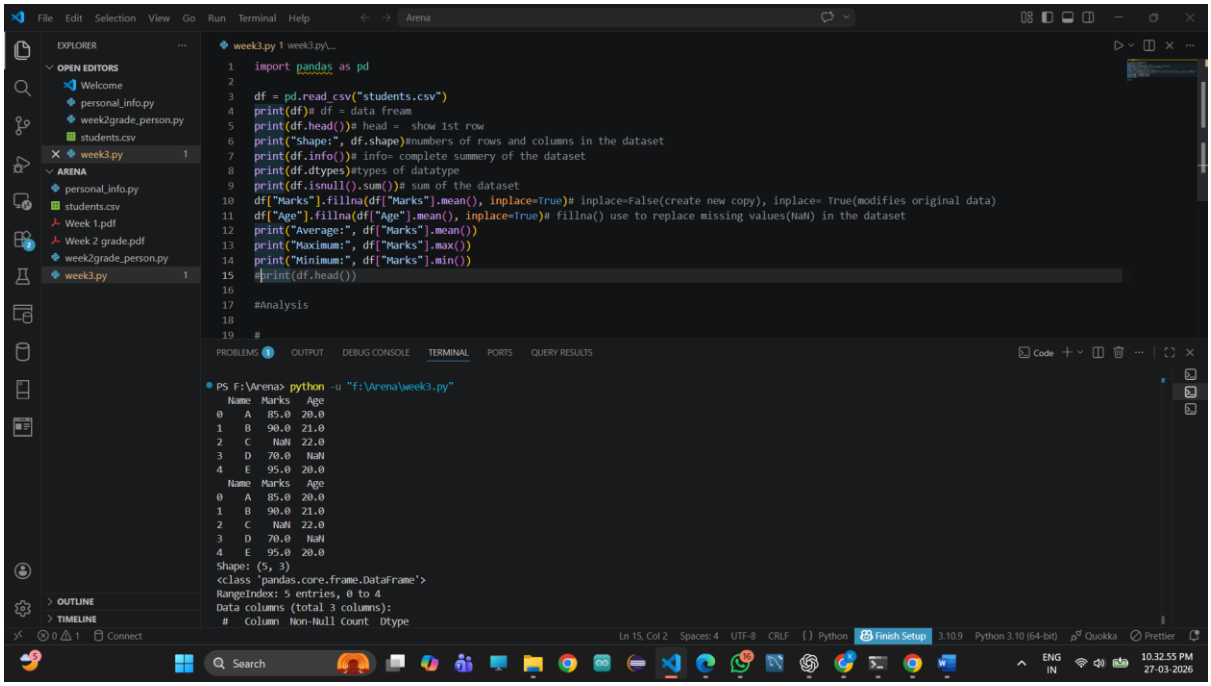
students.csv students.csv

```csv
Name,Marks,Age
A,85,20
B,90,21
C,,22
D,70,
E,95,20
```

sales.csv sales.csv

```csv
Product,Quantity,Price
Laptop,5,50000
Phone,10,20000
Tablet,3,15000
Phone,5,20000
Laptop,,50000
Tablet,4,15000
Headphones,8,3000
Monitor,2,12000
Keyboard,6,1500
Mouse,9,800
```

2.Explore Data - Check shape, columns, data types



3. clean data

week32.py 1 week32.py\...

```python
#Day 3: Clean Data
import pandas as pd

# Load dataset
df = pd.read_csv("sales.csv")

# Check missing values in each column
print("\nMissing Values:")
print(df.isnull().sum())

# Replace missing values in Quantity column with average
df["Quantity"].fillna(df["Quantity"].mean(), inplace=True)

# Remove duplicate rows if they exist
df.drop_duplicates(inplace=True)
# Show cleaned dataset
print("\nCleaned Dataset:")
print(df)
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS F:\Arena> python -u "f:\Arena\we
...
Cleaned Dataset:
      Product   Quantity  Price
0      Laptop   5.000000  50000
1       Phone  10.000000  20000
2      Tablet   3.000000  15000
3       Phone   5.000000  20000
4      Laptop   5.777778  50000
5      Tablet   4.000000  15000
6  Headphones   8.000000   3000
7     Monitor   2.000000  12000
8    Keyboard   6.000000   1500
9       Mouse   9.000000    800
```

4.Anayze sales

```python
df["Revenue"] = df["Quantity"] * df["Price"]

# Show dataset with revenue column
print("\nDataset with Revenue:")
print(df)

# Calculate total revenue
total_sales = df["Revenue"].sum()

# Calculate average price
average_price = df["Price"].mean()

# Find highest price
highest_price = df["Price"].max()

# Find lowest price
lowest_price = df["Price"].min()

# Find best selling product
# groupby() groups data by product
best_product = df.groupby("Product")["Quantity"].sum().idxmax()
```

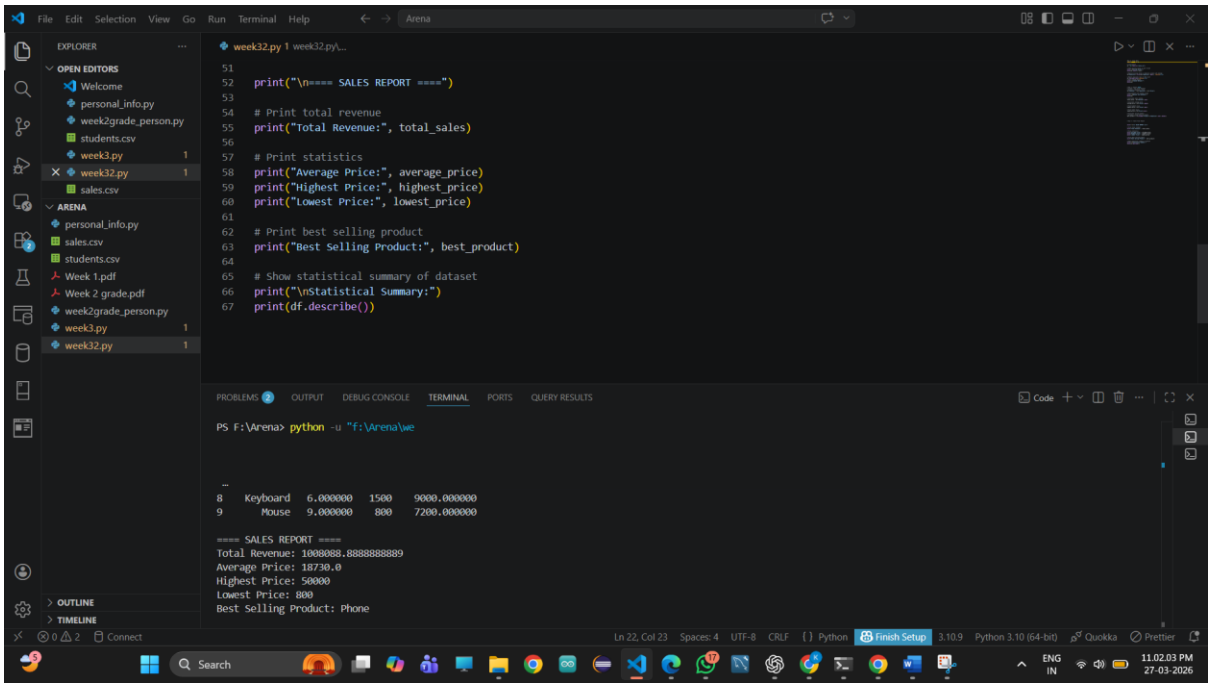
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS F:\Arena> python -u "f:\Arena\we



 …
Dataset with Revenue:
      Product   Quantity  Price        Revenue
0      Laptop   5.000000  50000  250000.000000
1       Phone  10.000000  20000  200000.000000
2      Tablet   3.000000  15000   45000.000000
3       Phone   5.000000  20000  100000.000000
4      Laptop   5.777778  50000  288888.888889
5      Tablet   4.000000  15000   60000.000000
6  Headphones   8.000000   3000   24000.000000
7     Monitor   2.000000  12000   24000.000000
```

5. final report



Summary

```
PROBLEMS 2    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    QUERY RESULT

PS F:\Arena> python -u "f:\Arena\we



 ...

==== SALES REPORT ====
Total Revenue: 1008088.8888888889
Average Price: 18730.0
Highest Price: 50000
Lowest Price: 800
Best Selling Product: Phone

Statistical Summary:
        Quantity         Price        Revenue
count  10.000000     10.000000      10.000000
mean    5.777778  18730.000000  100808.888889
std     2.572408  17943.618488  106061.988863
min     2.000000    800.000000    7200.000000
25%     4.250000   5250.000000   24000.000000
50%     5.388889  15000.000000   52500.000000
75%     7.500000  20000.000000  175000.000000
max    10.000000  50000.000000  288888.888889
PS F:\Arena>
```